

Arya Zadebrahimi

Bioinformatics Scientist

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🌐 [Online Resume](#)

Professional Summary

A highly motivated Bioinformatics graduate with hands-on experience in computational biology and software development. Proficient in core NGS data analysis, including sequence alignment and differential gene expression, with practical knowledge of tools such as HISAT, Samtools, and IGV. Possesses a strong foundation in bioinformatics algorithms with a strong interest in solving complex biological problems. Eager to contribute to a dynamic team and further develop expertise as a bioinformatician.

Experience

Iran University of Medical Sciences Tehran, Iran
Intern Aug 2022

Completed hands-on training in molecular biology techniques, including cell lysis and DNA extraction, PCR-based detection of specific DNA sequences, cell culture, and microscopic cell counting.

Nobaar Tehran, Iran
Technical Lead Jun 2018 – Dec 2018

Developed two applications for customers and company drivers, and led the design and architecture of the application APIs, collaborating with the backend development team on their implementation.

Mibarim Tehran, Iran
Android Developer Oct 2017 – Mar 2018

Developed two applications supporting ridesharing services and a separate app for comparing transportation options across ridesharing platforms.

Education

Sharif University of Technology Master of Science • GPA 18.2
Bioinformatics Tehran, Iran • Sep 2021 – Jul 2024

- Developed a mathematical model for synthetic microbial community design, exploiting metabolic network topology.
- Gained hands-on experience with an NGS analysis toolkit, including HISAT2 for read alignment, SAMtools/BCFtools for post-alignment processing, and FASTA/FASTQ file handling.
- Strengthened computational skills through advanced coursework in computer science and multiple teaching assistantships.

Islamic Azad University Bachelor of Science • GPA 15.58
Software Engineering Tehran, Iran • Sep 2015 – May 2021

- Established a solid foundation in programming, databases, and statistics through software development and computational problem-solving coursework
- Acquired proficiency in multiple programming languages, rapidly mastering new technologies as needed (e.g., learned Julia within one week to deliver a constrained optimization solution)
- Engineered an open-source Android application using Bluetooth Low Energy (BLE) to enable message sending, receiving, and broadcasting

Skills

🧬 Bioinformatics Analysis

Sequence alignment, Differential gene expression analysis, Pathway and Gene Ontology (GO) analysis, Motif finding / motif discovery algorithms, Phylogenetic tree construction, scRNA expression Analysis, FASTA/FASTQ file processing, Mathematical modeling, Constrained optimization, Molecular docking

📦 Bioinformatics Tools & Softwares

HISAT2, Samtools, BFCtools, Beagle, SHAPEIT, IGV (Integrative Genomics Viewer), Guassian, AutoDock Vina, CellProfiler, VMD

</> AI, Coding

Python, Pytorch, R / Limma • Bioconductor, LLMs, Image Processing, Julia, ReactJS, NodeJS, Android, Java, TCL

🗄️ Databases

NCBI, UniProt, PDB, ChEMBL, PubChem, BiGG, MongoDB, MySQL

🧪 Laboratory Techniques

DNA extraction, PCR, Cell culture